

SC-SCOPE scope decision: all-orders feasible for $I \leq 1\text{e-}3$, second-cumulant accepted at the $I=2\text{e-}3$ endpoint

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

This note records the operator-authorized decision (2026-06-07) on the SC-SCOPE all-orders endpoint, and consolidates the endpoint arc into one place. SC-SCOPE is the named second-cumulant hypothesis of the B1 Reading-H T6 selection; the "all-orders lift" is the separate question of whether the third-cumulant corrections preserve the selection at every certified intensity. This note does not change any tier.

2. The arc (what was established)

- **M-ENDPOINT** (scscope-mendpoint-evaluation v1.1): the endpoint dressing variance $M(0.33675) = 0.10495$ evaluated directly (cross-check 0.61%); the sunset-axis endpoint is $\times 1.13 > 1$ dressed (the U4 $\times 0.97$ was a frozen-coupling artefact).
- **GHAT4-PERTRANSFER** (scscope-endpoint-joint-assessment v1.0): the per-transfer quartic-difference form factor $\widehat{G^4}(t) = (J*J)(t)$ on the realized chords; convention-free shape reduction 0.64; quartic-difference alone $\times 1.29 > 1$.
- **Joint endpoint** (same note): the individually-positive channels (2nd $\times 2.60$, sunset $\times 1.13$, quartic $\times 1.29$, tadpole 0) JOINTLY over-consume the layer margin by $\times 1.32 \Rightarrow$ additive joint $\times 0.757 < 1$.
- **Joint incompatible-pairing** (scscope-joint-pairing v1.0): the most-favourable pairing recovers $\times 0.757 \rightarrow \times 0.905$ but not unity; the sunset ALONE is $\times 1.076$ (near-saturating). Per-transfer/pairing refinements are exhausted.

3. The intensity profile and the decision

The endpoint obstruction is confined to the thinnest intensity. With the recomputed second-order floors $\rho = (\times 59.4, \times 8.8, \times 2.6)$ at $I = (4 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3})$ and the paired joint inflation $1 + 1.87 = 2.87$ (endpoint), the all-orders paired joint ratio is

$$I = 4 \times 10^{-4} : \times 20.7, \quad I = 10^{-3} : \times 3.1, \quad I = 2 \times 10^{-3} : \times 0.905.$$

The all-orders third-order lift is thus FEASIBLE for $I \leq 10^{-3}$ and obstructed only at the $I = 2 \times 10^{-3}$ endpoint.

Operator decision (2026-06-07): accept second-cumulant scope at the $I = 2 \times 10^{-3}$ endpoint. The all-orders lift is recorded as feasible for $I \leq 10^{-3}$; the thinnest endpoint is selected at second-cumulant order. This is consistent with – and already priced into – the B1 T6 statement, which holds at matched second-cumulant order BY THE SC-SCOPE HYPOTHESIS. No tier changes; the two non-per-transfer routes (a sharper STEP-5B endpoint floor $\rho \gtrsim 3.9$; or a tighter third-cumulant treatment) remain available should full all-orders endpoint closure be desired later.

4. Resulting status

- B1-RH-ENUM: T6 CONDITIONAL on {H-LAYER, H-ADM-COH, SC-SCOPE} – UN-CHANGED. SC-SCOPE remains the named second-cumulant hypothesis.
- SC-SCOPE all-orders lift: FEASIBLE for $I \leq 10^{-3}$; second-cumulant ACCEPTED at the $I = 2 \times 10^{-3}$ endpoint (operator decision). The endpoint is no longer an OPEN research action – it is a recorded scope.
- M-ENDPOINT: RESOLVED. GHAT4-PERTRANSFER: per-transfer evaluated. ROBUSTNESS-MU2: CLOSED@[$\times 0.5, \times 2$]-2ND-CUMULANT (earlier this session).

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, 6 code-discipline)

- (α) “*Accepting scope is hiding a failure.*” DISMISSED: the failure is fully documented (NG-2026-06-07-scscope-endpoint-joint; the four arc notes) with the exact obstruction ($\times 0.905$, sunset near-saturating) and the named routes to close it. Accepting scope is a recorded decision with a reopening path, not a concealment.
- (β) “*Does the endpoint acceptance weaken B1 T6?*” DISMISSED: B1 T6 is second-cumulant BY HYPOTHESIS (SC-SCOPE); the endpoint acceptance is the hypothesis, not a new concession.
- (γ) “*Is the $I \leq 1e-3$ feasibility over-claimed?*” VALID-with-mitigation: it rests on the cited R_{sup} and the sunset/quartic estimates; it is recorded as FEASIBLE (estimate-grade), not PROVED. The endpoint negative is the robust part.

Quantitative sanity check (mandatory): magnitude – the paired joint ratios $\times 20.7 / \times 3.1 / \times 0.905$ track the second-order floors $\times 59.4 / \times 8.8 / \times 2.6$ divided by the common inflation 2.87; sign-direction – all third-order channels are costs; limit-case – at $I \leq 10^{-3}$ the floor is large enough that the inflation cannot break it, consistent with the feasibility statement.

6. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / SC-SCOPE scope decision (U16)
Precise statement:	Operator-authorized (2026-06-07): SC-SCOPE all-orders third-order lift FEASIBLE for $I \leq 1e-3$ (paired joint $\times 3.1$ at $1e-3$, $\times 20.7$ at $4e-4$); second-cumulant scope ACCEPTED at the $I=2e-3$ endpoint (paired joint $\times 0.905$, not closable by per-transfer/pairing). Capstone of M-ENDPOINT/GHAT4/joint/pairing.
Scope:	SC-SCOPE hypothesis of B1 T6; decision record.
Dependencies:	scscope-mendpoint-evaluation v1.1; scscope-endpoint-

	joint-assessment v1.0; scscope-joint-pairing v1.0; NG-2026-06-07-scscope-endpoint-joint.
Evidence grade:	DECISION RECORD over EXECUTED evidence (4 scripts, all asserts pass).
Reproduction command:	python codes/vacuum/scscope_joint_pairing.py
Expected output:	paired endpoint x0.905; feasibility for $I \leq 1e-3$.
Falsification gate:	A STEP-5B endpoint floor $\rho > \sim 3.9$ at $I = 2e-3$ or a tighter third cumulant (would CLOSE the endpoint all-orders and supersede the scope acceptance).
Tier before / after:	No tier change. B1 T6 unchanged. SC-SCOPE: all-orders FEASIBLE $I \leq 1e-3$; 2nd-cumulant ACCEPTED at $I = 2e-3$ endpoint (recorded scope, not an open action).
No-overclaim statement:	The endpoint is NOT all-orders closed; second-cumulant scope is accepted there by operator decision. $I \leq 1e-3$ feasibility is estimate-grade. B1 T6 selection unaffected (second-cumulant by hypothesis).
Next required action:	None required for SC-SCOPE (scope accepted). Optional reopening: sharper STEP-5B endpoint floor. Mainline moves to B1 ESTIMATOR-UPGRADE / other sectors.